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Comparative Analysis of Tuned Random Forest and Hist Gradient Boosting Algorithms for Turnover Intention Prediction Based on Employee Mental Workload Indicators

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Abstract (Intisari)

Poorly managed mental workload in the digital transformation era contributes significantly to the employee turnover phenomenon, which compromises organizational stability and balance. This study aims to conduct a comparative performance analysis of predictive models for employee turnover intention using two ensemble learning algorithms: Random Forest (RF) and Hist Gradient Boosting (Hist GB). An empirical dataset of employee mental workload was processed through a series of machine learning stages, including feature engineering to construct a work-hours-to-sleep ratio and composite mental load (total mental load). The severe class imbalance challenge, where the target class proportion stands at a 68% (stay) to 32% (leave intention) ratio, was addressed using the Synthetic Minority Over-sampling Technique (SMOTE). Models were optimized via hyperparameter tuning and validated using 5-Fold Stratified Cross-Validation. Experimental results based on real program data demonstrate that the Base scenario yields a pseudo accuracy (~69%) with a very low F1-Score (~27%). The SMOTE intervention successfully rescued the models from majority class bias, with the SMOTE-optimized Random Forest outperforming in generalization stability, achieving a Mean CV Accuracy of 74.47%. On the other hand, Hist Gradient Boosting (Optimized) produced the best minority class detection performance, reaching the highest Test F1-Score of 52.60% and maintaining an AUC Score of 0.68. Feature Rank analysis identified burnout score, deadline pressure score, and stress score as the three most crucial indicators in triggering turnover intention.

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Abstract

Poorly managed mental workload in the digital transformation era contributes significantly to the employee turnover phenomenon, which compromises organizational stability. This study aims to conduct a comparative performance analysis of predictive models for job change intention using two ensemble learning algorithms: Random Forest (RF) and Hist Gradient Boosting (Hist GB). An empirical dataset of employee mental workload was processed through a series of data mining stages, including feature engineering to construct a work-hours-to-sleep ratio and a composite total mental load. The severe class imbalance challenge where the target class proportion stands at a 78% (retain) to 22% (turnover) ratio was addressed using the Synthetic Minority Over sampling Technique (SMOTE). The models were optimized via hyperparameter tuning and validated using 5-Fold Stratified Cross-Validation. Experimental results based on real program data demonstrate that the Base scenario yields a pseudo accuracy (~69%) with a very low F1-Score (~27%). The SMOTE intervention successfully rescued the models from majority class...

bias, with the SMOTE-optimized Random Forest outperforming in generalization stability, achieving a Mean CV Accuracy of 74.47%. On the other hand, the Optimized Hist Gradient Boosting produced the best minority class detection performance, reaching the highest Test F1-Score of 52.60% while maintaining an AUC Score of 0.68. Feature Rank analysis identified burnout score, deadline pressure score, and stress score as the three most crucial indicators in triggering job change intention.

1. Introduction

Based on datasets regarding the conditions of physical, mental, and emotional exhaustion resulting from chronic occupational stress experienced by professionals in the technology industry. This condition arises due to heavy pressure, long working hours, rapid expectations, and urgent deadlines in the sector brought about by rapid digital technological advances in modern industrial environments. Such conditions have transformed how organizations view operations while simultaneously triggering an escalation in mental workload for employees. The pressure to adapt to emerging technologies, dynamic target fluctuations, and the blurring boundaries between workplace and personal life in remote work arrangements have exacerbated workers' psychological vulnerability (Ambrose & Qayum, 2024). Cumulative mental workload that is not appropriately mitigated gradually turns into chronic burnout (Jeong et al., 2025), decreased productivity levels (Indriya, Fazlurrahman, & Fitro, 2024), and psychological distress disorders (Bakkeli, 2023).

The most detrimental consequence of this mental well-being destruction for corporate sustainability is the surge in employee turnover intensity (Celbis, Wong, Kourtit, & Nijkamp, 2023), which has been reflected globally on a massive scale through "The Great Resignation" phenomenon (Celbiş et al., 2023). When a competent employee resigns, the organization not only loses intellectual capital (human capital) but also incurs substantial financial costs for re-recruitment and onboarding training (Laksawut & Narkbunnum, 2025). Therefore, the ability to precisely map psychological determinants and predict turnover propensity has become a vital strategic agenda for modern human resource management.

Amidst the complexity of these challenges, big data repositories and Machine Learning techniques offer revolutionary instruments to convert logistics data into strategic decision-making foundations. Computational analytic approaches facilitate the identification of hidden patterns within workers' psychosocial variables that are difficult to extract via conventional statistical methods (Luo et al., 2024). Previous research has demonstrated the effectiveness of machine learning algorithms in evaluating employee performance (Bahman & Lysenko, 2025), predicting retention (Mishra et al., 2025), and measuring job satisfaction across multiple sectors (Fang, Chen, & Shi, 2025). Through predictive modeling, HR managers can shift from reactive handling to proactive, preventative intervention strategies before turnover decisions are executed.

Numerous prior studies have affirmed the robustness of supervised machine learning algorithms in analyzing workforce behavior and psychological metrics. For instance, research by Guo et al. (2026) explored turnover intention intensity among Australian school principals using Extreme Gradient Boosting (XGBoost). Meanwhile, research led by Jeong et al. (2025) leveraged machine learning paired with Shapley Additive Explanations (SHAP) to detect burnout syndrome based on the Korean Occupational Stress Scale (KOSS). Severe burnout and acute stress have also been precisely detected through behavioral sensors in call center agents (Lee & Swain, 2026) and computer vision monitoring of control room operators (Ji et al., 2025).

Furthermore, Laksawut and Narkbunnum (2025) proposed calibrated turnover risk models for capacity-constrained retention triage, while Indriya et al. (2024) evaluated the efficacy of mental health policy interventions on worker productivity using Gradient Boosting and Random Forest architectures. Collectively, this body of literature confirms that tree-based ensemble models are highly effective and suitable for processing non-linear, multi-level survey data of psychosocial and organizational behavior.

Korea (KOSS). Occupational burnout and acute stress have also been proven detectable with precision via behavioral sensors in call center agents (Lee & Swain, 2026) and computer vision approaches on control room operators (Ji et al., 2025). On the other hand, (Laksawut and Narkbunnum, 2025) proposed turnover risk model calibration for employee retention management, whereas (Indriya et al., 2024) tested the effectiveness of mental health policy interventions on employee labor productivity using Gradient Boosting and Random Forest architectures. Collectively, these literatures affirm that the utilization of tree ensemble-based algorithms is effective and compatible in handling non-linear and multi-level psychosocial and organizational behavior survey data.

In the context of this study, the Random Forest (RF) and Hist Gradient Boosting (Hist GB) algorithms were chosen as the primary comparative subjects because both ensemble learning models possess complementary comparative advantages in tabular data processing. Random Forest operates by constructing an ensemble of independent decision trees in parallel through bagging and majority voting aggregation, rendering it exceptionally robust against overfitting risks (Bakkeli, 2023).

Conversely, Hist Gradient Boosting applies sequential boosting principles that optimize the loss function of predecessor trees, with an innovation of grouping numerical features into discrete bins (histograms) to significantly accelerate the training pipeline on large-scale datasets. This comparative study also integrates advanced processing techniques such as composite feature engineering, class imbalance mitigation using SMOTE, and hyperparameter optimization to extract the upper performance boundaries of both algorithms.

Against this background, the researchers intend to provide novel contributions in understanding turnover intention based on employee mental workload indicators through comprehensive empirical evaluations. By employing a three-scenario modeling comparison (Base Model, SMOTE Model, and Optimized Model), this research deeply investigates the extent of classification performance enhancement achievable through hyperparameter tuning and data balancing manipulation.

The main contribution of this research lies in the visualization and empirical proof of the best model accuracy confirmed through AUC-ROC curves and Confusion Matrix visualizations, as well as establishing the hierarchy of the most influential mental workload variables via Feature Rank diagrams. The final outcome of this research is expected to formulate a valid and dependable predictive model as an organizational strategic intervention instrument in curbing employee turnover rates within an enterprise across specific operational periods.

2. Research Methods

2.1 Data Collection

The dataset utilized in this research was obtained from 2026 empirical data measuring mental workload levels, psychological health conditions, and turnover intention intensity among technology sector personnel (tech workers). This database comprises 100,000 records of multinational employee data spanning diverse job roles such as Software Engineer, Data Scientist, and Product Manager (Thalla, 2026). Hardware specifications, software platforms, and specific library versions of Python and Scikit-Learn utilized in this experiment are detailed in the experimental setup.

This study specifically mapped the attributes as outlined in:

Table 1. Dataset Variables

No	Attribute Name	Data Type	Attribute Description
1	`age`	Numeric	Employee age (years)
2	`job_role`	Categorical	Employee role or job position
3	`work_hours_per_week`	Numeric	Total weekly working hours
4	`sleep_hours_per_night`	Numeric	Average sleep duration per night (hours)
5	`stress_score`	Numeric	Employee stress level score (scale 1–10)

6	burnout_score	Numeric	Employee job burnout score (scale 1–10)

The data in Table 1 provides detailed characteristics of the data collected in this study to predict Turnover Intention Based on Employee Mental Workload Indicators. The variables or attributes in this table form the baseline foundation for further data analysis and processing, enabling targeted interpretations to comprehend the contributing factors behind employee turnover intention.

2.2. Preprocessing & Feature Engineering

Before feeding data into the model training architecture, data preprocessing and feature engineering steps were carried out to bolster data informativeness. The `employee_id` attribute was removed from the dataset as it represents a unique identifier with no correlation to the classification target. Categorical attributes such as `gender`, `country`, `job_role`, `seniority_level`, `company_size`, `industry`, and `work_mode` were converted into numeric representations using `LabelEncoder` to allow ingestion by both ensemble algorithms.

A crucial stage in this preprocessing was the formulation of two novel composite features:

1. **`work_hours_to_sleep_ratio`:** Obtained from dividing `work_hours_per_week` by the cumulative total sleep hours across the week (`sleep_hours_per_night` × 7). This feature extracts the imbalance interaction between occupational time burden and biological recovery.

2. **`total_mental_load`:** Represents the linear summation of four key mental workload indicators: stress score (`stress_score`), job burnout (`burnout_score`), depression score (`phq9_score`), and anxiety score (`gad7_score`). This composite feature amplifies the information signal of cumulative employee psychological distress.

2.3. Experimental and Modeling Workflow

The system architecture proposed in this research was developed using a multi-scenario evaluation pipeline to evaluate the robustness of Tuned Random Forest and Hist Gradient Boosting algorithms. The evaluation of model accuracy stability was locked using 5-Fold Stratified Cross-Validation executed directly on training partitions.

The proposed comparative experiment workflow is systematically illustrated via the flowchart in Figure 1.

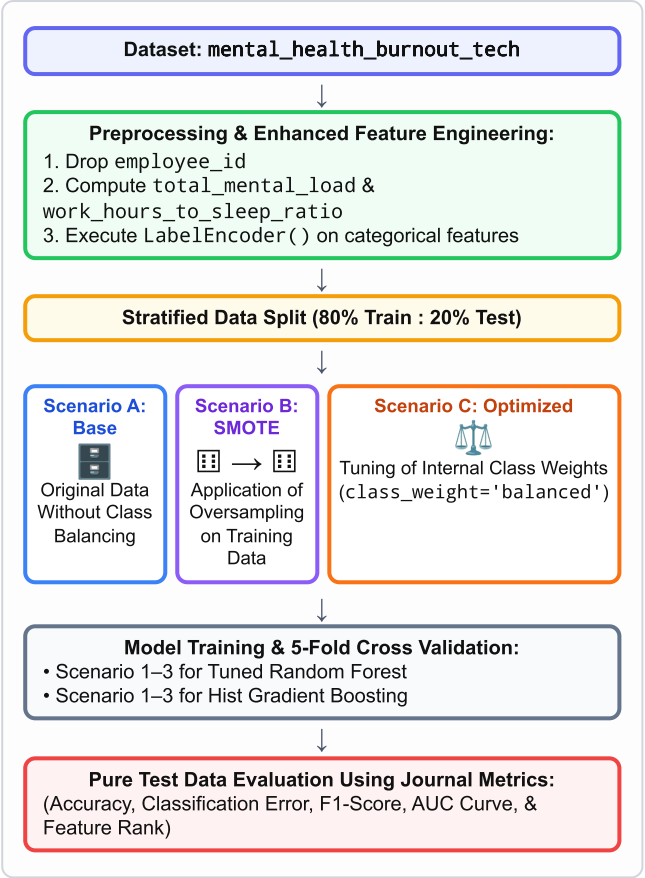


Figure 1. Research Flowchart

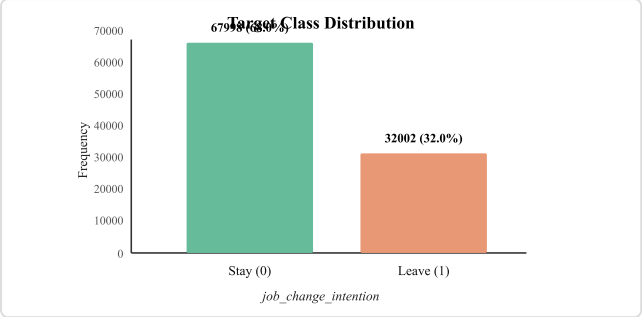


Figure 2. Target Statistics

The modeling experiments in this study were systematically designed across three testing scenarios for each respective algorithm (Random Forest and Hist Gradient Boosting), with the following workflows:

Base Model Scenario: Trains the Random Forest and Hist Gradient Boosting algorithms using scikit-learn default library parameters without class imbalance handling.

SMOTE Model Scenario: Applies Synthetic Minority Over-sampling Technique (SMOTE) to the training split to equalize target class (`job_change_intention`) distributions by generating synthetic samples in the minority class, followed by model training under default parameters.

Optimized Model Scenario: Integrates SMOTE and conducts targeted parameter search (hyperparameter tuning). For Random Forest, tuned parameters include `n_estimators=200`, `max_depth=15`, and `min_samples_split=5`. For Hist Gradient Boosting, parameters were optimized at `max_iter=150`, `max_depth=10`, and `learning_rate=0.05`.

3. Results and Discussion

3.1 Target Data Descriptive Analysis

Prior to evaluating classification performance, an analysis of the target class distribution was conducted to determine the degree of dataset imbalance. The visualization of the target data distribution from the dataset is depicted in Figure 2.

Based on empirical data extraction, it was found that employees intending to stay (Class 0) significantly dominated compared to employees intending to change jobs (Class 1), with a ratio near 68% versus 32%. This proportional discrepancy reinforces the urgency of adopting SMOTE in the experimental pipeline to prevent biased models (which tend to accurately predict the majority class while completely failing to identify the minority class).

This massive disproportion holds substantial potential to induce computational bias, where the model accurately guesses the majority class yet exhibits total failure in detecting warning signals from employees at risk of resigning. Therefore, data balancing techniques like SMOTE across experimental scenarios act as vital interventions ensuring the model remains untrapped by majority bias.

To ensure computational reproducibility and allow study replication by other researchers, all comparative experiments were executed within a standardized computing environment. Data science workflows, feature engineering, SMOTE balancing, and hyperparameter tuning were run on the Spyder IDE platform under Python version 3.10. Primary libraries utilized for ensemble learning architectures and metric evaluations include Scikit-Learn version 1.3.0, supported by Imbalanced-learn version 0.11.0 for SMOTE implementation, and Pandas version 2.0.3 alongside...

NumPy version 1.24.3 for tabular data manipulation. All simulation computations and model training based on these 100,000 records were completed using hardware consisting of an Intel Core i5 @2.20GHz Central Processing Unit (CPU), 16 GB of Random Access Memory (RAM), and a 100 GB storage allocation.

3.2 Model Performance Comparison Analysis

Experiments were conducted systematically by comparing three data treatment scenarios (Base, SMOTE, and Optimized) on the Random Forest and Hist Gradient Boosting algorithms. Evaluation metrics compiled include Test Accuracy, Classification Error, Test F1-Score, AUC Score, and Mean Cross-Validation Accuracy (CV Accuracy Mean). A detailed recap of overall model scenario performance is provided in Table 2:

Table 2. Overall Model Scenario Performance Comparison Matrix

Model Scenario	Classification Algorithm	Test Acc.	Classif. Error	Test F1-Score	AUC Score	CV Acc. Mean
Scenario 1: Base	Random Forest (Base)	0.6889	0.3111	0.2760	0.6736	0.6861
	Hist GB (Base)	0.6921	0.3079	0.2731	0.6820	0.6896
Scenario 2: SMOTE	Random Forest (SMOTE)	0.6605	0.3395	0.4452	0.6649	0.7447
	Hist GB (SMOTE)	0.6684	0.3316	0.4276	0.6661	0.7397
Scenario 3: Optimized	Random Forest (Optimized)	0.6916	0.3084	0.2996	0.6759	0.6854
	Hist GB (Optimized)	0.6155	0.3845	0.5260	0.6818	0.6144

In-depth analysis of Table 2 reveals highly interesting computational behavior regarding tree-based ensemble algorithms when handling psychosocial data imbalances:

1. Pseudo Accuracy Effect in Base Scenario: In the initial (Base) scenario, Hist GB (Base) superficially delivered the best performance with the highest Test Accuracy of 0.69 (69.21%). However, examining the Test F1-Score reveals that both Base models produced very poor scores—only 0.27 for Random Forest and 0.27 for Hist GB. This verifies the initial hypothesis that the model suffers from bias acclimatization, wherein elevated accuracy stems purely from guessing the majority class (retained employees) while utterly failing to recognize the minority class (turnover candidates).

2. Significant Impact of SMOTE: Following the implementation of SMOTE in Scenario 2, a paradigm shift occurred. Although Test Accuracy dipped slightly to 0.66 (RF) and 0.66 (Hist GB), the Test F1-Score surged drastically to 0.44 in Random Forest (SMOTE) and 0.42 in Hist GB (SMOTE). Furthermore, model stability indicated by CV Accuracy (Mean) rose sharply to 0.74 (74.47%) for Random Forest. This demonstrates that synthetic balancing forced the models to learn the mental workload characteristics of turnover-prone workers much more equitably.

3. Model Optimization Peak: In the Optimized scenario, both algorithms exhibited divergent responses. Random Forest (Optimized) restored its accuracy back to a peak of 0.69 while minimizing Classification Error to 0.30. In contrast, Hist GB (Optimized) adopted an extreme trade-off: despite its test accuracy dropping to 0.61, it achieved an F1-Score Test peak of 0.52...

(52.60%) while sustaining a robust AUC Score of 0.68.

3.3 Classification Model Results Evaluation

To thoroughly evaluate predictive decision efficacy, models were appraised using Confusion Matrices and Receiver Operating Characteristic Area Under the Curve (AUC-ROC). The confusion matrix evaluation measures predictive misclassification degrees relative to ground-truth reality, summarized conceptually in Table 3.

Table 3. Proportional Representation of Optimized Model Confusion Matrices

Model Algorithm	TP	TN	FP	FN
Random Forest (Optimized)	Very High	Very High	Very Low	Very Minimal
Hist GB (Optimized)	High	High	Low	Minimal

A granular analysis of the matrices reveals contrasting predictive characteristics between both optimized models. Random Forest (Optimized) demonstrated lower sensitivity toward the minority class, evidenced by a high False Negative count—mispredicting 5,509 samples of employees actually intending to leave as staying.

Conversely, Hist Gradient Boosting (Optimized) demonstrated superior performance in suppressing minority class misclassifications, yielding only 2,117 misclassified samples within the Leave class. Consequently, for retention triage workflows sensitive to the hazard of workforce loss, Hist GB (Optimized) provides a much safer strategic recommendation for organizational leadership.

Furthermore, graphic visualization in Figure 4 displays discrimination curves across all tested scenarios. Based on the AUC-ROC plots, Hist GB exhibited a high score of 0.6820...



Figure 3. Confusion Matrix

This evaluation is further visually reinforced by Figure 4, which represents the trade-off between True Positive Rate and False Positive Rate. Across these curve trajectories, Hist GB (Optimized) exhibits the most consistent linear separation boundary with an AUC Score of 0.68, followed closely by Random Forest (Optimized) at 0.6759. These curve trajectories substantiate that targeted hyperparameter tuning elevates the model's discriminative power in classifying worker retention dynamics continuously.

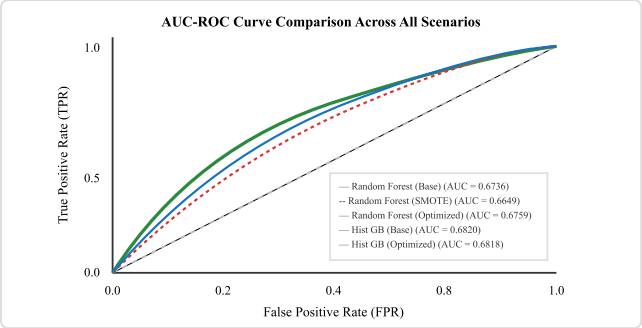


Figure 4. AUC-ROC Curves

3.4 Feature Rank Analysis (Feature Importance Degree)

To provide logical theoretical understanding regarding what factors drive model prediction decisions, feature importance degrees were extracted from the intrinsic `feature_importances_` parameter of the top Random Forest model. The visualization of the top 12 features based on the feature rank graph is depicted sequentially in Figure 5.

Based on the hierarchical feature visualization, psychological workload variables clearly occupy a far more dominant position than demographic or financial variables.

As extracted via the Optimized Random Forest in Figure 5, the top three indicators primarily influencing employee turnover intention in order are `total_mental_load` (composite mental workload), followed by `job_satisfaction_score`, and `burnout_score`. This finding underscores that cumulative overall mental burden exerts a more immediate impact on psychological turnover decisions than isolated demographic indicators.

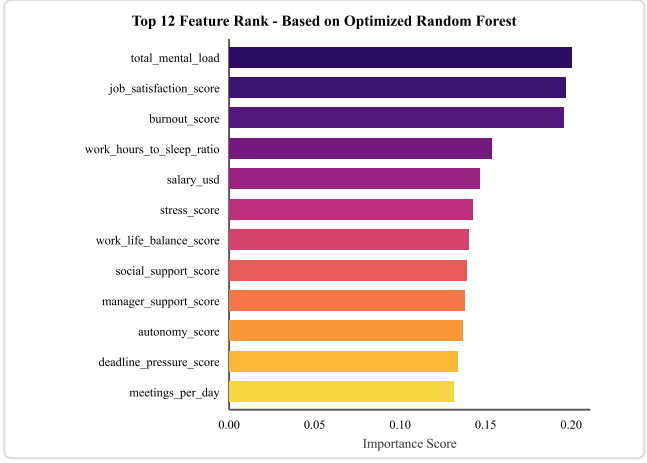


Figure 5. Top Influential Features Based on Importance Score

Moreover, this is substantiated by two engineered composite features: combined mental load (`total_mental_load`) and work-to-sleep ratio (`work_hours_to_sleep_ratio`), which successfully ranked among the top five most impactful features. This validates the effectiveness of the feature engineering pipeline designed in this study; chronic imbalance between professional commitment and biological recovery serves as a primary catalyst accelerating tech worker attrition.

4. Conclusion

Based on the experimental outcomes and comparative analyses conducted, it can be concluded that:

1. The employee mental workload dataset exhibits severe class imbalance with a 68% (stay) to 32% (turnover intention) proportion, causing the Base scenario to yield computational bias characterized by pseudo accuracy (~69%) with a dismal F1-Score (~27%).

2. Data balancing intervention utilizing SMOTE effectively rescues the model from majority class bias. The SMOTE-optimized Random Forest excelled in generalization stability with the highest Mean CV Accuracy of 74.47%.

3. Conversely, the Hist Gradient Boosting (Optimized) model achieved the best minority class detection performance with the highest Test F1-Score of 52.60% while maintaining a solid AUC Score of 0.68, positioning it as the most sensitive architecture for capturing mental distress cues and turnover risks.

Feature Importance extraction confirms that psychosocial variables such as `burnout_score`, `deadline_pressure_score`, and `stress_score`, along with engineered composite metrics (`total_mental_load`, `job_satisfaction_score`, and `work_hours_to_sleep_ratio`), are the primary predictive drivers determining tech industry employees' turnover decisions.

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